SeedHammer II — the taproot pathological wallet

The hardest wallet this constellation describes, round-tripped on the machine.

This is the four-tier degrading vault — the wallet `CORPUS.md` calls *the* pathological example — expressed as a **depth-3 taproot tree**:

```
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,{and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c39
8023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,@0/48'/0'/0'/2'/<0;1>/*,@1/48'/0'/1'/2'/<0;1>/*,@2/48'/0'/2'/2'/<
0;1>/*))),{and_v(v:after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_
a(2,@3/48'/0'/3'/2'/<0;1>/*,@4/48'/0'/0'/2'/<0;1>/*,@5/48'/0'/1'/2'/<0;1>/*))),{and_v(v:older(65535),multi_a(2,@6/48'/0'/2'/
2'/<0;1>/*,@7/48'/0'/3'/2'/<0;1>/*)),and_v(v:older(4255898),multi_a(1,@8/48'/0'/0'/2'/<0;1>/*,@9/48'/0'/1'/2'/<0;1>/*,@10/4
8'/0'/2'/2'/<0;1>/*))}}})
```

Two `sha256` hashlocks. Both Bitcoin timelock flavours, absolute and relative. `multi_a` at thresholds 3, 2, 2 and 1. A **NUMS internal key**, so there is no key-path spend at all — every route to the coins goes through a script leaf. Eleven cosigners across **three masters**.

> **Until three commits ago this document could not exist.** Every one of its four leaves is `and_v(v:…)` wrapping a timelock or a hashlock, and the device's tap-leaf reader named only `pk` / `multi_a` / `sortedmulti_a`. It returned *zero* leaves and the consent screen read "Complex policy — display only", while the host derived the same wallet's addresses without complaint. That was F-214; the fix replaced the vocabulary with an emitter, and this journey is the proof it worked.

> The keys are BIP-39 test vectors. Never put funds behind them.

1 · Every slot's origin matches its own key file

The template carries eleven per-key origins, written by hand; the eleven key files declare them independently. If the two disagreed, the card would name a path the key does not live at — and **nothing downstream would notice**, because addresses come from the xpub a card carries, never from the origin it declares. That is F-217's lesson, and this gate is the answer to it.

```
  @0  48'/0'/0'/2' ok
  @1  48'/0'/1'/2' ok
  @2  48'/0'/2'/2' ok
  @3  48'/0'/3'/2' ok
  @4  48'/0'/0'/2' ok
  @5  48'/0'/1'/2' ok
  @6  48'/0'/2'/2' ok
  @7  48'/0'/3'/2' ok
  @8  48'/0'/0'/2' ok
  @9  48'/0'/1'/2' ok
  @10 48'/0'/2'/2' ok

0 slot(s) disagree with their key file
```

2 · The engraved artifact

The keyless template set — 4 chunks. No `--path`: the origins ride in the template, and `--path` would flatten eleven distinct origins onto one.

> **And nothing would stop it.**
> An earlier draft of this page said `md encode` "refuses outright". Measured: it refuses the *keyed* form, where it can see one fingerprint bound to several different keys — but on this *keyless* template `--path` exits **0** and produces a perfectly good card with eleven wrong origins. The check needs a fingerprint *and* an xpub per slot, and a keyless template carries neither (F-221). **The card that gets engraved is the form the guard cannot see**, so leaving `--path` off here is a choice this journey makes, not one the tool enforces.

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md encode tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d54
7bfee9ace803ac0,{and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),mul
ti_a(3,@0/48'/0'/0'/2'/<0;1>/*,@1/48'/0'/1'/2'/<0;1>/*,@2/48'/0'/2'/2'/<0;1>/*))),{and_v(v:after(1893456000),and_v(v:sha256
(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,@3/48'/0'/3'/2'/<0;1>/*,@4/48'/0'/0'/2'/<0;1>/
*,@5/48'/0'/1'/2'/<0;1>/*))),{and_v(v:older(65535),multi_a(2,@6/48'/0'/2'/2'/<0;1>/*,@7/48'/0'/3'/2'/<0;1>/*)),and_v(v:older
(4255898),multi_a(1,@8/48'/0'/0'/2'/<0;1>/*,@9/48'/0'/1'/2'/<0;1>/*,@10/48'/0'/2'/2'/<0;1>/*))}}}) --group-size 0 --force-ch
unked
chunk-set-id: 0x3d896
md1f8kykps9zjtvyyy5jmpprjjtvyy49ykcgfw2fdsssjjtvyyw2fdssj55jmpp9ef9kzzz2fdsturhn8pf8jq0c
md1f8kykpsgs3ef9kzz2j5qqcr32v6mqq85yszv6a4pxuusyh2unu8xqz8naa2r2akwukvgm42glxskfar6x4423
md1f8kykpssws7xkx9k7c9xu6pzkjqsspy9fntdcdhkyqfntk5ymnjqjatj0sucqg70h4gdtkec5muezu6y3f47m
md1f8kykpsmje3rw4fp6rc6cckmmq5mngy26gpzx3g4xdwqqq8lluszze6v6uqpq0px3qq2y6qqmxcdqfn702jwt
note: stdout is a keyless descriptor template (no keys)
[exit 0]

The engraved set is 4 md1 chunk(s).
```

3 · E1 — the cards re-encode to the policy they came from

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md verify --template tr(50929b74c1a04954b78b4b6035e97a5e078a5a
0f28ec96d547bfee9ace803ac0,{and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b
9a08ad),multi_a(3,@0/48'/0'/0'/2'/<0;1>/*,@1/48'/0'/1'/2'/<0;1>/*,@2/48'/0'/2'/2'/<0;1>/*))),{and_v(v:after(1893456000),and_
v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,@3/48'/0'/3'/2'/<0;1>/*,@4/48'/0'/0'/
2'/<0;1>/*,@5/48'/0'/1'/2'/<0;1>/*))),{and_v(v:older(65535),multi_a(2,@6/48'/0'/2'/2'/<0;1>/*,@7/48'/0'/3'/2'/<0;1>/*)),and_
v(v:older(4255898),multi_a(1,@8/48'/0'/0'/2'/<0;1>/*,@9/48'/0'/1'/2'/<0;1>/*,@10/48'/0'/2'/2'/<0;1>/*))}}}) md1f8kykps9zjtvy
yy5jmpprjjtvyy49ykcgfw2fdsssjjtvyyw2fdssj55jmpp9ef9kzzz2fdsturhn8pf8jq0c md1f8kykpsgs3ef9kzz2j5qqcr32v6mqq85yszv6a4pxuusyh2u
nu8xqz8naa2r2akwukvgm42glxskfar6x4423 md1f8kykpssws7xkx9k7c9xu6pzkjqsspy9fntdcdhkyqfntk5ymnjqjatj0sucqg70h4gdtkec5muezu6y3f4
7m md1f8kykpsmje3rw4fp6rc6cckmmq5mngy26gpzx3g4xdwqqq8lluszze6v6uqpq0px3qq2y6qqmxcdqfn702jwt
OK
[exit 0]
```

Why `md verify` and not a `md decode` diff. Decode renders the per-key origins *away*: the text it returns is well-formed, missing the one field a signer needs to find its keys, and re-encodes to a *different card*. A journey asserting decode-equality would either fail on a perfectly correct plate or be quietly weakened until it passed. `md verify` exists for exactly this question, returns 0 or 1, and compares payloads rather than rendered text. The gap is filed as F-219.

4 · The bind — both card sets are the same wallet

The engraved set is keyless; the set the device gathers carries the eleven xpubs. Without this assertion the journey could engrave one wallet and prove another. The *template* id is key-stable, which is precisely what makes it the right thing to compare.

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md inspect md1f8kykps9zjtvyyy5jmpprjjtvyy49ykcgfw2fdsssjjtvyyw
2fdssj55jmpp9ef9kzzz2fdsturhn8pf8jq0c md1f8kykpsgs3ef9kzz2j5qqcr32v6mqq85yszv6a4pxuusyh2unu8xqz8naa2r2akwukvgm42glxskfar6x44
23 md1f8kykpssws7xkx9k7c9xu6pzkjqsspy9fntdcdhkyqfntk5ymnjqjatj0sucqg70h4gdtkec5muezu6y3f47m md1f8kykpsmje3rw4fp6rc6cckmmq5mn
gy26gpzx3g4xdwqqq8lluszze6v6uqpq0px3qq2y6qqmxcdqfn702jwt
template: tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,{and_v(v:after(1000000),and_v(v:sha256(a84dce4
0975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*))),{and_v(v:after(1893
456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,@3/<0;1>/*,@4/<0;1>/*,@5/
<0;1>/*))),{and_v(v:older(65535),multi_a(2,@6/<0;1>/*,@7/<0;1>/*)),and_v(v:older(4255898),multi_a(1,@8/<0;1>/*,@9/<0;1>/*,@1
0/<0;1>/*))}}})
n: 11
wallet-policy-mode: false
md1-encoding-id: 3d89678ac64eef08b84652f4fcaf59ea
wallet-descriptor-template-id: 44ad26a19b53048b6ff8957359a30c31
origins:
  @0: m/48'/0'/0'/2'
  @1: m/48'/0'/1'/2'
  @2: m/48'/0'/2'/2'
  @3: m/48'/0'/3'/2'
  @4: m/48'/0'/0'/2'
  @5: m/48'/0'/1'/2'
  @6: m/48'/0'/2'/2'
  @7: m/48'/0'/3'/2'
  @8: m/48'/0'/0'/2'
  @9: m/48'/0'/1'/2'
  @10: m/48'/0'/2'/2'
wallet-policy-id: e5a97a4e1d45d28a19a21861da70c57e
wallet-policy-id-fingerprint: 0xe5a97a4e
note: stdout is a keyless descriptor template (no keys)
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md inspect md1fydqpts9zjtvyyy5jmpprjjtvyy49ykcgfw2fdsssjjtvyyw
2fdssj55jmpp9ef9kzzz2g9dxek2y7nujt5 md1fydqptsdkzz89ykcgf22sqrqw9fntvqq7sjqfntk5ymnjqjatj0sucqg70h4gdtkemje3zptvqjvcnjxwzs m
d1fydqpts464y8g0rtrzm0vznwdq3tfqggqjz5e4kuxmmzqye4m2zdeeqfw4e8cwvqy08m65gwj7tsrdu4lxh md1fydqptsc6hdnh9nzxa2jr583433dhkpfhxs
g45szydz32v6uqqq0lleqy9n5e4cqzq7zdz2p56enkuv0wmd md1fydqpt3qqz3xsvuvqu79mg9pw0za5z3883w6pgmnchdq5jux3r03twrg3hckhp5gmutms62a
dsg9szjhkkm md1fydqpt3g3hcc9pj9qp5jsezsq63gv3gqv9k6zqhwsv0jskp2rsal4egz4ep5859p875x67k9jqdtfexpz6x md1fydqpt3sxjz9m806prlsm7
94tkxqs6806lhaeh6reknylagmwyjycf8044xtt9flsdlkv2cdfwmwvyvtshs md1fydqpt3lf8tpwunu53n9svfgll7zvw5qnkxgeal5wa4wy86389z3gq870g9
n6rjwuq7cms9xujhgpxdl943 md1fydqptjqjfp99zcyrd98kjcu8vgdtu6gt04upy3z0n8ek4aj5kk9satjt7uv4am7jn7q2q8zxlcml2eczwj md1fydqptjdj
ly68gp3qxnzrvqpghktcljzr7j8mxp87s87gwfzjsys8njyze0uj4852ajv5x9h80ejfs5f0ev md1fydqptjneynsul328vme3ent5fcakske788nfhcc7nza0y
nr9p5d3exan7jvup3kx8742qsqc0mpk87fkuj md1fydqptjmfhtfgqfec6yprg0dc3gdls899kx6k89kfkaz2az67ajl8hyy967je7sks39z8we5emvhw6cqrza
md1fydqptnxmlcrstt5m3s998x2rk7av9gmew82jq5zap9302ynhp37ggd6z5u4emag0zr8gkdy3qw763syq0m md1fydqptnvhsxwt3qdt2f9njdjzxjf3pk2mw
xeza6dduqqnvtpl3vsgcj7vs58sl2hmlrnys76x2a7hhfx8n5 md1fydqptns67znc3fa47zw97hmwk2vvhq2h6g9y9evngtkrskaxskvp4qchq5qpv0pez9razz
nwk32ujcyjk5 md1fydqptnanu69tqlftc3g6hj0npmnk06drm5awv08c2v0vktw5t6qu9g4xkmv2dywrwkud7euvmc9egs7fl5 md1fydqpt5xgnjc4hh02k9m3
a3x9a4h733cpg59kdh7306na4nhq32cd5w9c2uzlhxryrtzduftjem8mym63un md1fydqpt5fgpxkjsesxnnp7e00y36klmrkvqhu772hljrwlhzlncxfd22zx6
72chhdu5ajtcmtv0vz8hye793 md1fydqpt5hd0zup072vfeww0h2t65xl3djg9z7s576y0hwv6a92khwlwfmwulc9zrqupz2z7sktxueuey8thn md1fydqpt5l
737tj9hrdzaved93u3a2nawefnqlk8ycusz0quhfev0a3hqdghjmksz3ry92dsr5uuvw3jqhu9r md1fydqpt49pjhxf00s49ugex8rue307wdslgcj5r8x9t5j5
4jfadzzykqz37sefnc0rhk5zckt9spyhvvhr3h md1fydqpt42amshp87fzcadzx07vs02d3t8efq2e50d8mg50ts8t8mquqwc90uynctuc4sv9q4ytx0xzzyq4c
k md1fydqpt45jnl3rgupw7m2jr5caw7vxt9v7rn6wz3pd0pwa4xqjj8e7htel64394h6gme2g5uavyj00qjctyx md1fydqpt4a8ajc9sqw28jr665c4hq9t4ts
w75nz845y9up7y276lt29ywu3usza3aqrp2qqj37hvham46nhy
template: tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,{and_v(v:after(1000000),and_v(v:sha256(a84dce4
```

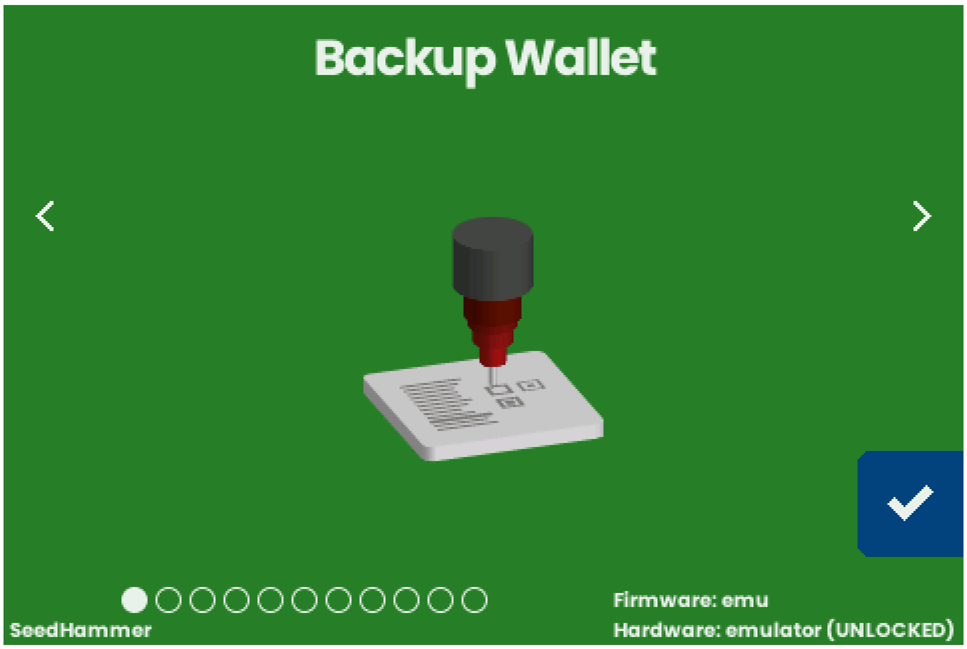
```
0975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*))),{and_v(v:after(1893
456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,@3/<0;1>/*,@4/<0;1>/*,@5/
<0;1>/*))),{and_v(v:older(65535),multi_a(2,@6/<0;1>/*,@7/<0;1>/*)),and_v(v:older(4255898),multi_a(1,@8/<0;1>/*,@9/<0;1>/*,@1
0/<0;1>/*))}}})
n: 11
… clipped; full text in transcript_tr_pathological.txt
```

5 · What the device must prove to

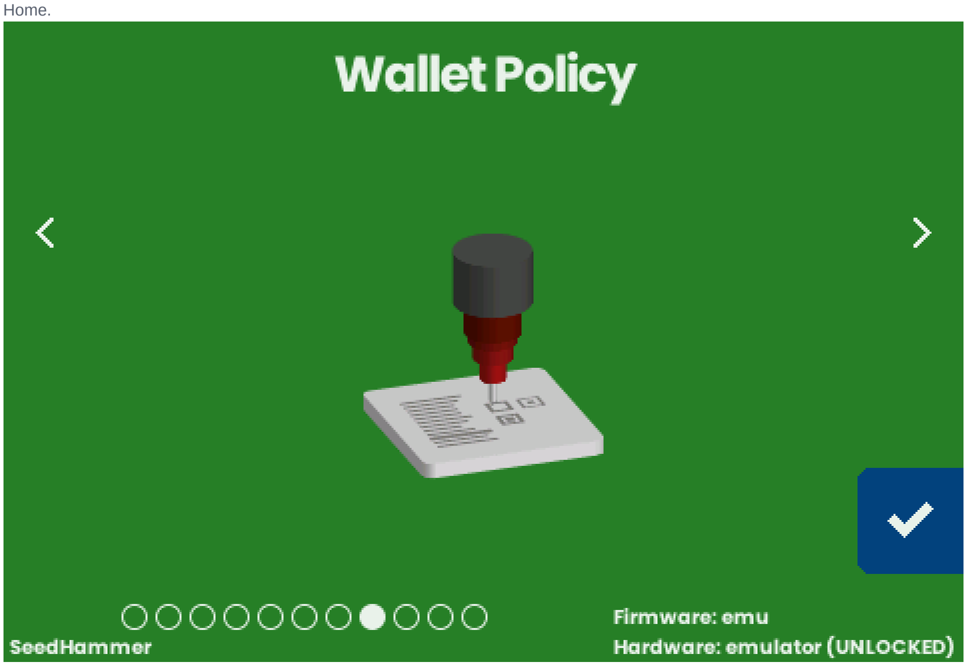
Derived from the *cards*, not from the argument list that made them.

wallet id`wallet-policy-id: 590f3abcaad2aca5a3f526917f5bb57a`
receive 0..1`bc1pgavpejq7pkax6zdtuay9ejenangza2t3l6gfka79h94p23xcexssak7pzr`
`bc1p3wy7tcypvkm49wtyeq2yw4t0266vxdnqlkny5nsgzguc0fcvdl3s5hkkp5`
change 0..1`bc1pnjrzcky5qqlhyz4rvhq5r96ca77wyccnfls0p5l7k8ae7402vjhshf3zya`
`bc1p7yrpv9q6j58vwh5yrnnj8ky6dnykn7z2lak4j0ckt52j84j8fxzs4h6khc`

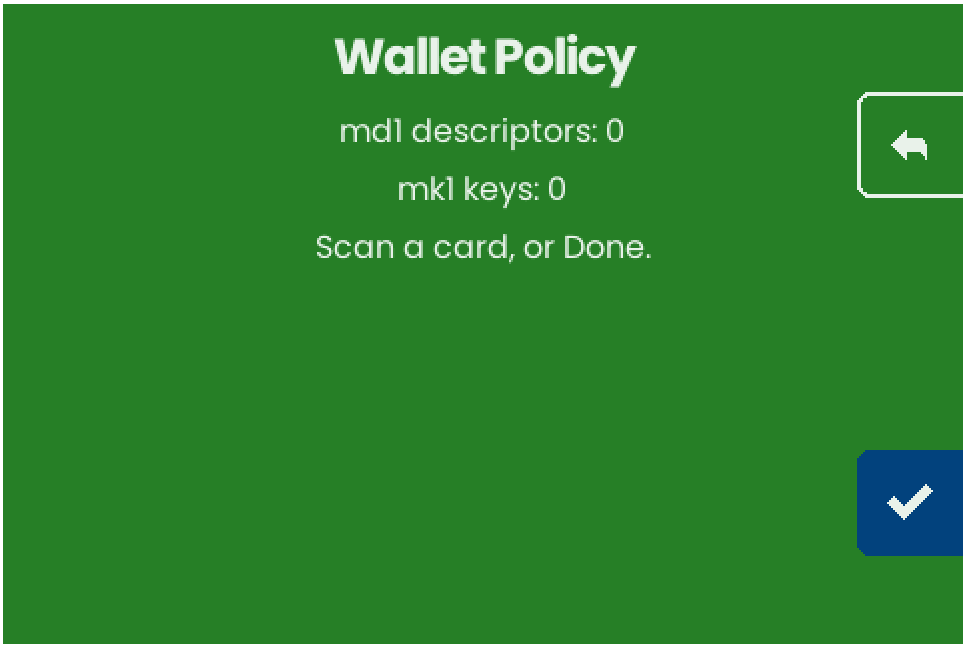
6 · The device



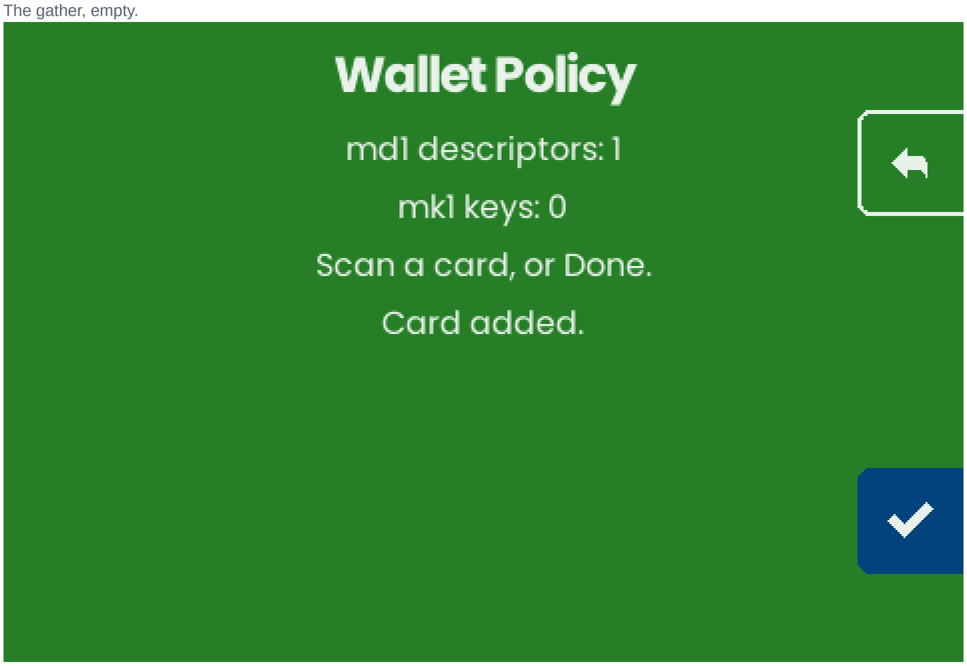
Home.



Wallet Policy.



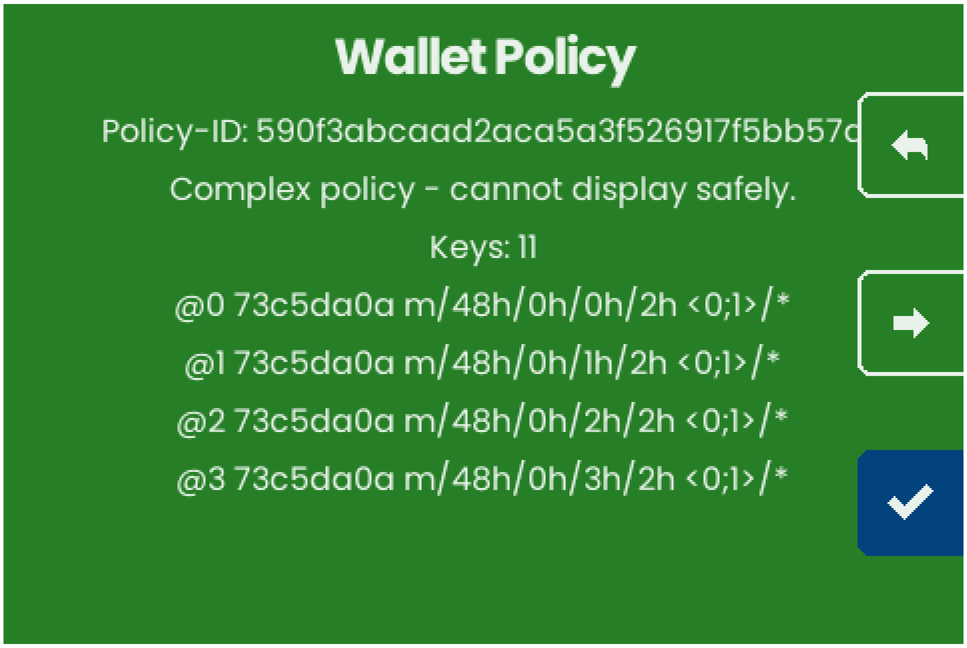
The gather, empty.



After all 24 chunks cross the reader: ONE card.

7 · The proof, on the consent path

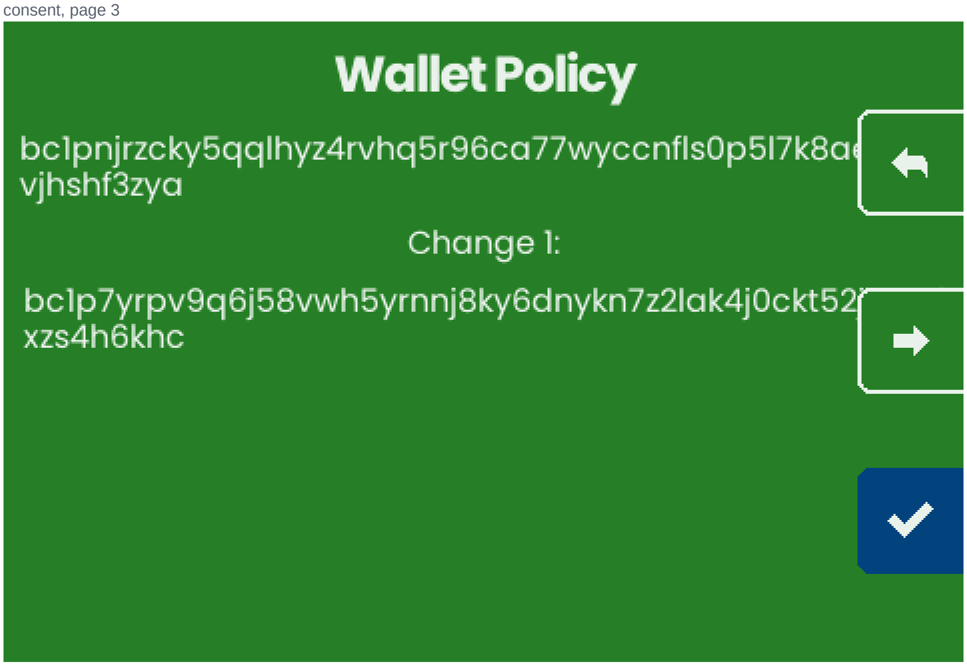
Eleven key slots print before the addresses do, so the screen pages 4 times — and the walk reads every page, because a bound tuned for a four-key wallet would stop before reaching the half that matters.



consent, page 1

consent, page 2

consent, page 3



consent, page 4

8 · What this run established

The capture is a **comparison**. It is handed the host's id and addresses and throws if the screen disagrees, and the absence assertions reject "display only" and "no addresses" outright — so the device cannot pass by refusing politely.

chunks presented24
cards gathered1
consent pages read4
wallet id matched`590f3abcaad2aca5a3f526917f5bb57a`
addresses matched`bc1pgavpejq7pkax6zdtuay9ejenangza2t3l6gfka79h94p23xcexssak7pzr`
`bc1p3wy7tcypvkm49wtyeq2yw4t0266vxdnqlkny5nsgzguc0fcvdl3s5hkkp5`
`bc1pnjrzcky5qqlhyz4rvhq5r96ca77wyccnfls0p5l7k8ae7402vjhshf3zya`
`bc1p7yrpv9q6j58vwh5yrnnj8ky6dnykn7z2lak4j0ckt52j84j8fxzs4h6khc`

The negative control is a command, not a memory. `python3 capture_tr_pathological.py --prove-it-can-fail` corrupts one character of one expected address and exits 0 only if the walk *caught* it. A comparison nobody has made fail is not evidence that it can.

Reproducing

```
bash transcript_tr_pathological.sh > transcript_tr_pathological.txt 2>&1
python3 capture_tr_pathological.py                    # rebuilds emu.wasm, drives it
python3 capture_tr_pathological.py --prove-it-can-fail # the negative control
python3 build_pdf_tr_pathological.py                  # this document
```

What this journey does NOT show

- **No spend, and for tiers 1–2 no spend is possible from this backup at all.** Both hashlock tiers need the 32-byte preimage to spend, and *zero* backup strings carry it (F-132).
- **The tiers do not degrade in the order they read.** Tier 4 (1-of-3) matures at 365 days; tier 3 (2-of-2) at ≈455 — the weakest key-set unlocks ~90 days *first* (F-133).
- **No plate read-back.** There is no reader; the card-set equality above is the honest proxy.
- **No ms1 secret leg.** Taproot-independent, and already shown in the earlier journeys.
- **Decode's output still does not round-trip** (F-219). The card carries its origins and both `md inspect` and `md decode` now report them — an earlier draft of this line said neither did, which its own section 4 already disproved. What remains is that `decode`'s *stdout* renders them away, so re-encoding what it prints yields a different card.